

- (e) (i) It is suggested that the quantities T , S and h are related by the equation

$$T^2 = A \frac{S}{h} + B$$

where A and B are constants.

Using your answers in (d)(iii), determine the values of A and B .
Give appropriate units.

$$A = \dots\dots\dots$$

$$B = \dots\dots\dots$$

[2]

- (ii) Theory suggests that

$$A = \frac{28\pi^2 L}{5g}$$

where g is the acceleration of free fall.

Use your values in (a) and (e)(i) to determine a value for g .
Give an appropriate unit.

$$g = \dots\dots\dots [1]$$

[Total: 20]

